Sumanth Rao

Ph.D. Candidate, Computer Science · University of California, San Diego
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Research Interests

Empirical security, Internet measurement, email and messaging security, network security, AI security.

Skills

Python, C++, Rust · Zeek, Wireshark, ZMap, Censys · Ghidra, JADX · Prompt-injection, Cyber ranges.

Education

2021 – exp. 2027 **Ph.D. Computer Science** — **University of California, San Diego**
Advisors: Stefan Savage and Geoffrey M. Voelker

2016 – 2020 **B.Tech. Computer Science** — **PES University, Bengaluru, India**
Advisor: Gowri Srinivasa · First Class with Honors · GPA 9.6/10

Experience

Jun – Sep 2026 **Google, Gmail Delivery Security** · *Student Researcher* — Mentor: Weihaw Chaung
– Built an observatory joining WHOIS and pDNS to catch SPF hijacks in 200M domains daily.

Jun – Sep 2025 **Corelight Labs** · *Research Intern* — Mentors: Yacin Nadji, Vern Paxson
– Built Zeek detection of DCSync/DCShadow attacks by fingerprinting DRSUAPI replication.

Jun – Sep 2022 **Google, Virtual Cluster** · *Software Engineering Intern*
– Reduced web-proxying overhead 7× in Google's sandboxed mock production environment.

2020 – 2021 **Citrix, Licensing** · *Software Engineer I*
– Cut License Server test resource use by up to 50% with Docker microservices on Xen.

2019; 2022 – 26 **Zulip Open Source Project** · *Volunteer Contributor* — Google Summer of Code · Mentor: Tim Abbott
– Developed the terminal client (GSoC 2019) and later mentored five GSoC students.

Selected Publications

Lost in Translation: Text Message Spoofing via Email – ***Distinguished Paper Award***
Sumanth Rao, Ye Shu, Stefan Savage, Aaron Schulman, Geoffrey M. Voelker, Enze Liu. *IEEE S&P (Oakland), 2026.*

Patient Care Technology Disruptions Associated With the CrowdStrike Outage
Jeffrey L. Tully*, **Sumanth Rao***, Isabel Straw, Rodney A. Gabriel, Chris Longhurst, Stefan Savage, Geoffrey M. Voelker, Christian J. Dameff. *JAMA Network Open, 2025.* *equal contribution

Unfiltered: Measuring Cloud-based Email Filtering Bypasses
Sumanth Rao, Enze Liu, Grant Ho, Geoffrey M. Voelker, Stefan Savage. *ACM Web Conference (WWW), 2024.*

BLE Theft Auto: Evaluating the Security of Aftermarket BLE-based Automotive Remote Control Systems
Jerry Yu*, Yibo Wei*, **Sumanth Rao**, Mohak Vaswani, Jefferson Chien, Christian Dameff, Nishant Bhaskar, Aaron Schulman. *USENIX Security, 2026.*

No Privacy Among Spies: Assessing the Functionality and Insecurity of Consumer Android Spyware Apps
Enze Liu, **Sumanth Rao**, Sam Havron, Grant Ho, Stefan Savage, Geoffrey M. Voelker, Damon McCoy. *PETS, 2023.*

Achievements

– **Distinguished Paper Award**, IEEE Symposium on Security and Privacy (Oakland). 2026
– CSE Doctoral Award for Excellence in Service and Leadership, UC San Diego. 2026
– Finalist, CSAW Applied Research Competition. 2026
– Top 5, BSides San Diego CTF (2025); Finalist, Stanford DEFCON NatSec Hackathon (2024). 2024 – 2025
– JN Tata Endowment Scholar (national graduate scholarship). 2021 – 2022
– CVE-2025-46311 (Apple iOS), CVE-2026-18411 (CISA); talks FTC PrivacyCon.
– Provisional U.S. Patent 63/674,158, "Digital Availability Surveillance Method". 2024
– **Program Committee**: ACM SIGMETRICS 2027, IEEE EuroS&P 2027, RAID 2026.